

Readings of pronouns across connectives are sensitive to monotonicity¹

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Abstract. Kanazawa (1994) proposed a generalization connecting the monotonicity properties of quantifiers and the preferred reading (existential or universal) of donkey sentences. We show through two experimental studies that his generalization extends to cross-conjunction anaphora. In a nutshell, we find that simple cross-conjunction anaphora like *there is a circle and it is blue* are read existentially (*there is a blue circle*) but adding negation to the second conjunct (*there is a circle and it is not blue*) makes an additional universal reading accessible (*there is a circle and no circle is blue*). These results provide a challenge to Egli's principle, a principle validated by most existing dynamic theories, according to which only an existential reading should be found in these constructions, regardless of the presence of negation.

Keywords: pronoun, donkey, cross-conjunction anaphora, Kanazawa's generalization, Egli's principle.

1. Introduction

The so-called donkey sentences, illustrated in (1), have been a major challenge to theories of pronouns. In these sentences, a pronoun may co-vary with an indefinite antecedent, even when the antecedent does not c-command the pronoun.

- (1) Donkey Sentences (Geach, 1964)
- a. Every farmer who owns a donkey pats it.
 - b. No farmer who owns a donkey pats it.
 - c. Some farmer who owns a donkey pats it.

Deriving the possibility of co-varying readings in such non-c-commanding configurations is a hard problem that has often required sophisticated analyses (Heim, 1982; Kamp et al., 2011; Evans, 1980, 1977; Cooper, 1979; van der Does, 1993). Even when a particular analysis accounts for the potential of co-variation, there is still the issue of which truth-conditions should be predicted. Two primary types of readings have been discussed: the existential reading, as in (2a), and the universal reading, as in (2b).

- (2) **Paraphrase:** *every farmer who owns a donkey cherishes...*
- a. ... *at least one of the donkeys they own* (existential reading)
 - b. ... *all of the donkeys they own* (universal reading)

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Very little is known for certain regarding how these truth-conditions come about and it is even debated whether the paraphrases in (2) constitute readings in the proper sense of the word (Champollion et al., 2019). It is observed that there is typically a preference for one reading. When the quantifier is *every* as in (1a), the reading obtained is typically universal. On the other hand, donkey sentences headed by *some* and *no*, in (1c) and (1b), are overwhelmingly interpreted existentially as in (2a).

In a classical paper (Kanazawa 1994), Kanazawa proposed a generalization linking the interpretations accessed most naturally to the monotonicity properties of the quantifier that heads the sentence (i.e. *every* in the case of (1)). He proposed that when a quantifier’s restrictor and scope are of opposite monotonicity, the universal reading is preferred; when the scope and the restrictor are of the same monotonicity, it is the existential reading that is preferred.

The goal of this work is to investigate readings of pronouns in a different construction: cross-conjunction anaphora, illustrated in (3) and (4). In this construction too, a pronoun co-varies with an indefinite antecedent although the latter does not c-command the former. There too, it is an empirical question whether these sentences receive an existential reading (as in (3a) or (4a)), a universal reading (as in (3b) or (4b)). The main question addressed by this work is: is there an effect of monotonicity on cross-conjunction anaphora? Does the generalization established by Kanazawa (1994) extend to cross-conjunction anaphora?

- (3) Sam owns a donkey_i and it_i is gray.
 - a. existential: *Sam owns a donkey and some donkey he owns is gray.*
 - b. universal: *Sam owns a donkey and every donkey he owns is gray.*
- (4) Sam owns a donkey_i and it_i isn’t gray.
 - a. existential: *Sam owns a donkey and some donkey he owns isn’t gray.*
 - b. universal: *Sam owns a donkey and every donkey he owns isn’t gray.*

There are two theoretical reasons to expect that Kanazawa’s generalization does not extend to these constructions. First, Dynamic Semantics, one of the main traditions in the semantics of pronouns, typically validates a principle, which we call Egli’s principle following Elliott (2020). According to this principle, formally expressed in (5), the reading of cross-conjunction anaphora is the same as if the indefinite were to take scope over the entire conjunction.

- (5) **Egli’s principle:**

$$[\exists x, p] \wedge q \Leftrightarrow \exists x, [p \wedge q]$$

If this principle is correct, (3) and (4) should be equivalent to (6a) and (6b) respectively, which are alternative paraphrases of the existential reading. The presence of negation should therefore not affect the observed reading.

- (6) a. Sam owns a donkey that’s gray.
- b. Sam owns a donkey that’s not gray.

Second, some accounts of the existential/universal ambiguity (Champollion et al., 2019; Chierchia, 2009) explicitly tie it to the presence of a quantifier and thus to the donkey construction. They therefore do not expect Kanazawa’s generalization to go beyond the donkey cases.

Defying these expectations, we report the result of two experimental studies showing that

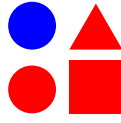


Figure 1: Example of display for a trial

Kanazawa's generalization does in fact extend to cross-conjunction cases. More specifically, we find that when the pronoun and its antecedent stand in environments of same monotonicity, as in (3), only the existential reading in (3a) is observed. This is consistent with Egli's theorem. However, when negation is added to the second conjunct, both an existential and a universal reading seem possible. This is exactly what Kanazawa's generalization expects, as we show, but it is inconsistent with Egli's principle. Our results therefore place an important challenge to theories of Dynamic Semantics.

The roadmap is as follows. In section 2, we present the result of our first study comparing the readings accessed in (3) and in (4). In section 3, we address a potential objection according to which our results might simply be due to well-known processing difficulties associated with negation, by means of a second experimental study. Section 4 concludes.

2. Experiment I: readings of pronouns across conjunctions

(This study was pre-registered at <https://osf.io/afqm4>.)

2.1. Methods

This study is a truth-value judgment task. In each trial, participants are presented with a sentence and a picture containing four colored shapes (two rows of two shapes), as in fig. 1. They are asked to rate each trial on a 7-point scale, ranging from "completely false" to "completely true".

There are three types of sentences, given in (7), varying the names of the shape or shapes mentioned (square, circle and triangle) and the colors mentioned (blue, red, green). The POS sentences are of the form in (7a). They display a cross-conjunction anaphora and don't involve negation. The NEG sentences are of the form in (7b) and include a negation in the second conjunct. The CONTROL sentences, of the form in (7c), don't contain any cross-conjunction anaphora and are used as attention checks.

- | | | | |
|-----|----|---|-----------|
| (7) | a. | There is a circle and it is blue. | (POS) |
| | b. | There is a circle and it isn't blue. | (NEG) |
| | c. | There is a circle and the square is blue. | (CONTROL) |

We seek to find whether our participants access the existential and universal readings of POS and NEG sentences paraphrased in (8) and (9). Each type of sentence is paired with four different types of images. These image types are designed to discriminate between the different readings. The resulting 8 conditions for the target POS and NEG sentences (4 conditions per sentence type) are described in Tables 1 and 2, including the predicted truth-value of the sentence depending on the reading.

- | | |
|-----|--|
| (8) | There is a circle _i and it _i is blue. |
| a. | existential: <i>There is a circle and some circle is blue.</i> |

- b. universal: *There is a circle and every circle is blue.*
- (9) There is a circle_i and it_i isn't blue.
- a. existential: *There is a circle and some circle isn't blue.*
- b. universal: *There is a circle and every circle isn't blue.*

Table 1: Experimental conditions for positive sentences

Example picture	Description	Condition Name	$\exists$ -reading	$\forall$ -reading
	no circle	Pos-F1 st	F	F
	1 non-blue circle	Pos-F2 nd	F	F
	1 blue circle, 1 non-blue circle	Pos- $\exists$	T	F
	2 blue circles	Pos- $\forall$	T	T

Table 2: Experimental conditions for negative sentences

Example picture	Description	Condition Name	$\exists$ -reading	$\forall$ -reading
	no circle	Neg-F1 st	F	F
	1 blue circle	Neg-F2 nd	F	F
	1 blue circle, 1 non-blue circle	Neg- $\exists$	T	F
	2 non-blue circles	Neg- $\forall$	T	T

The CONTROL sentences are paired with two types of pictures. In the CONTROL-TRUE condition, the picture contains a single circle and a single square and the unique square is blue. In the CONTROL-BOTHFALSE, the picture contains no circle, just a single square but the square is not blue.

In summary, there are 10 conditions: 4 picture types for POS sentences, 4 picture types for NEG sentences and two picture types for CONTROL sentences. Each condition is presented three times, resulting in 30 trials in total. The order of the trials is completely randomized.

2.2. Participants

The study recruited 120 participants using the Prolific platform. Each participant completed all 30 trials. We excluded participants who reported being color-blind or not being native speakers of English. We also excluded participants who failed to assign one of the two lowest scores (1 and 2) to the CONTROL-FALSE condition in at least two of the trials, and participants who did not assign one of the two highest scores (6 and 7) to the CONTROL-TRUE control condition in at least two trials. Based on these criteria, 6 participants were excluded from the final analysis.

2.3. Results

Fig. 2 shows the mean ratings per condition.

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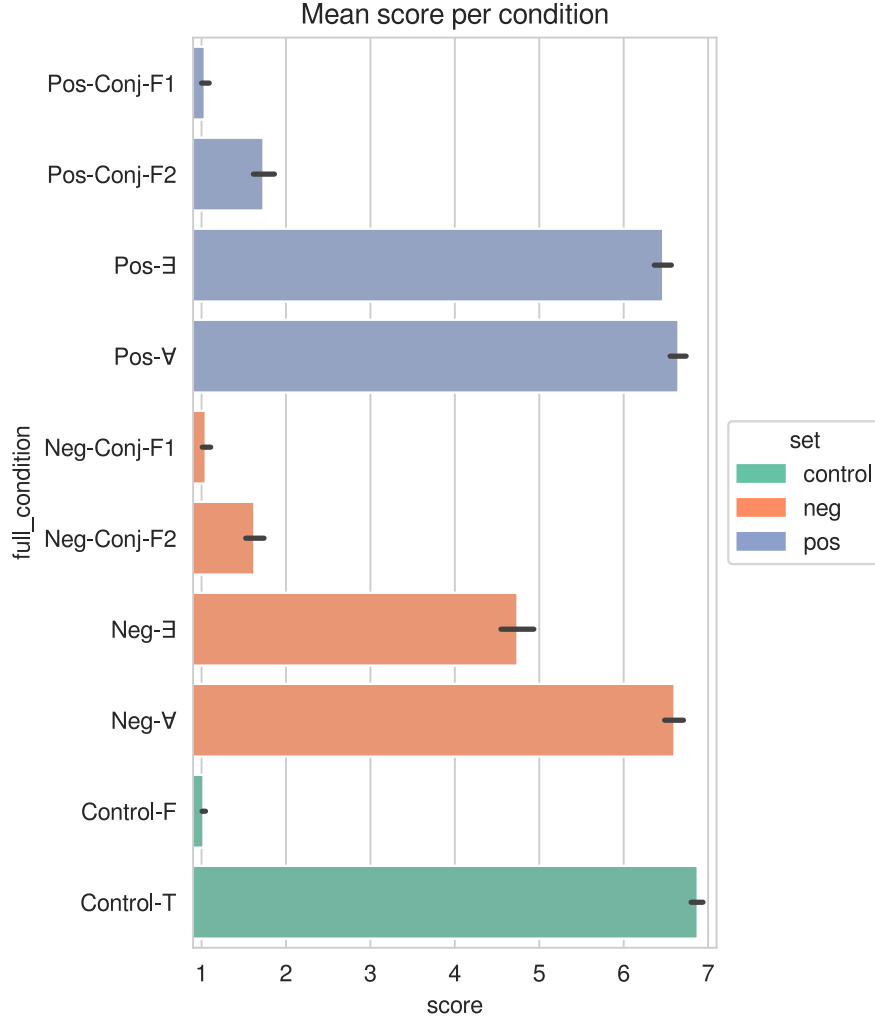


Figure 2: Mean rating per conditions

Every pairwise comparison reported below was conducted using the following method: first, we extracted the data for the conditions to be compared; then, we fitted a mixed-effects CLM model to these data with the effect to be tested as a regressor and a random intercept per participant; next, we fitted another model with a fixed intercept and a random intercept per participant but without the effect to be tested; finally, we conducted likelihood ratio tests to compare the deviance of the two models.

We also tested the interaction between Pos/Neg and $\exists/\forall$. We extracted the data from the four relevant conditions and fitted a full model that included both factors (Pos/Neg and $\exists/\forall$) and their product as regressors, as well as an additive model without the interaction term. In both cases, we included a random intercept for participants. The results of these comparisons are reported with their corresponding χ^2 and p-values. To account for multiple comparisons, the Holm-Bonferroni correction was applied; consequently, the p-values reported below are corrected.

The difference between *Pos-F2nd* and *Pos- $\exists$* was significant ($\chi^2(1) = 798.91, p < 2.2 \times 10^{-16}$).

The difference between $Pos-\exists$ and $Pos-\forall$ was also significant ($\chi^2(1) = 17.031$, $p = 3.679 \times 10^{-5}$). The difference between $Neg-F2nd$ and $Neg-\exists$ was significant ($\chi^2(1) = 441.01$, $p < 2.2 \times 10^{-16}$), as was the difference between $Neg-\exists$ and $Neg-\forall$ ($\chi^2(1) = 301.5$, $p < 2.2 \times 10^{-16}$).

Notably, the interaction between Pos/Neg and $\exists/\forall$ was also significant ($\chi^2(1) = 105.37$, $p < 2.2 \times 10^{-16}$).

2.4. Discussion

Overall, there seems to be, across POS and NEG sentences, a clear cline whereby POS/NEG-F2nd conditions are rated lower than POS/NEG- $\exists$ conditions and the latter POS/NEG- $\exists$ are rated lower than POS/NEG- $\forall$.

A first possible interpretation of these facts would be that POS and NEG sentences receive both an existential and a universal reading. Furthermore, since there is a significant interaction between POS/NEG and $\exists/\forall$, this would entail that the universal reading (represented by the difference between $\exists$ and $\forall$ conditions) is much more accessible in the NEG sentences than in the POS sentences.

But a different interpretation is suggested by the precise distribution of responses, which can be visualized through the histogram of responses to the POS/NEG- $\exists/\forall$ conditions (fig. 3). We find that, for POS sentences, both the $\exists$ and the $\forall$ conditions have a similar distribution of responses: a unimodal distribution peaked at around 7. But there is a stark contrast for the NEG conditions: while the NEG $\forall$ shows the same distribution of responses as POS- $\exists$ and POS- $\forall$, the NEG- $\exists$ elicits many more intermediate responses (around 4), as well as falsity judgments (around 1).

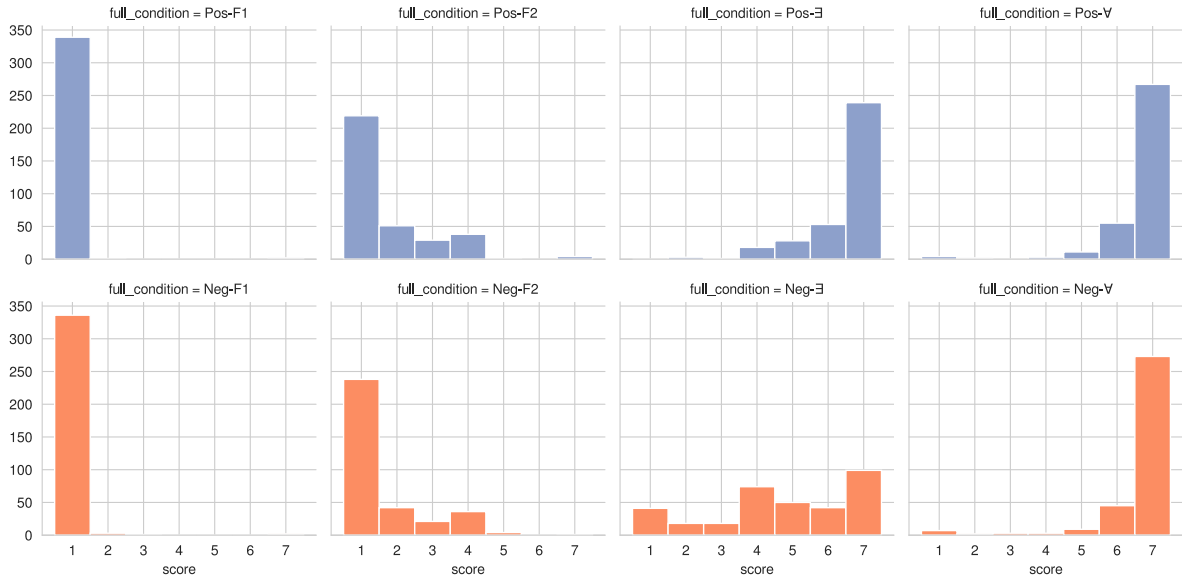


Figure 3: Histogram of responses for POS and NEG conditions

This suggests an alternative interpretation of the differences found between the conditions: the difference between POS- $\exists$ and POS- $\forall$ is not due to the presence of a universal reading but to a truth-gradiency effect. While POS- $\exists$ and POS- $\forall$ are both true under an existential reading, one of the conditions is closer to situations of falsity than the other. There is evidence that

The study recruited 119 participants using the Prolific platform. Each participant engaged in all 30 trials, consisting of 3 trials across 10 different conditions (5 conditions per target sentences, and 2 conditions per control sentences). As in experiment I, we excluded participants

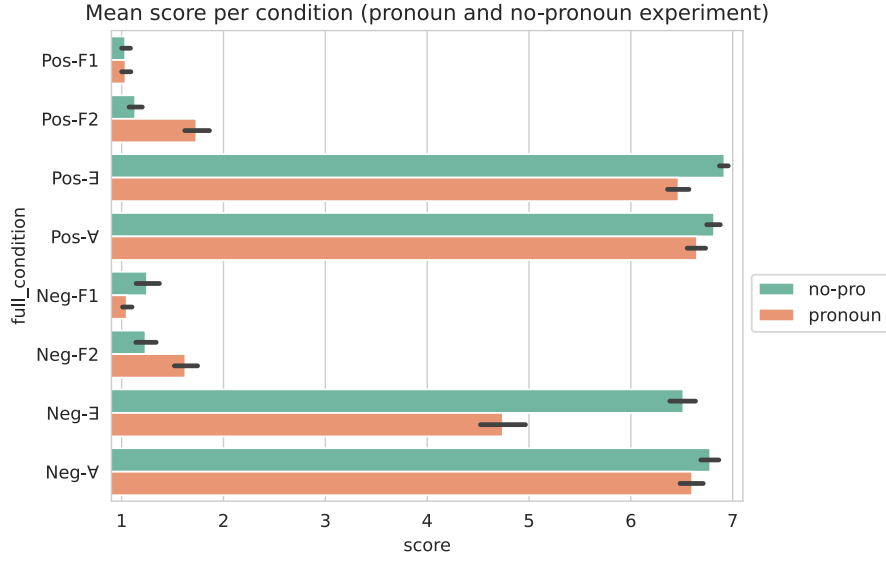


Figure 4: Bar plot of mean scores for all conditions across experiment I (pronoun) and experiment II (no-pronoun)

who reported being color-blind or not being native speakers of English. We likewise excluded participants who failed to assign one of the two lowest scores (1 and 2) to the CONTROL-FALSE condition in at least two of the trials, and participants who did not assign one of the two highest scores (6 and 7) to the CONTROL-TRUE control condition in at least two trials. Based on these criteria, 4 participants were excluded from the final analysis.

3.3. Results and discussion

The mean score obtained for each condition are given in fig. 4, where they are put side-by-side with the mean score of the corresponding condition in experiment I. One salient difference between the results of the two experiments is the difference between the NEG-PRO- $\exists$ and the NEG-NOPRO- $\exists$ (cf fig. 5): the $\exists$ picture appears much less acceptable when the sentence contains a negation and a pronoun. This difference translates into a significant interaction between POS/NEG and PRO/NOPRO when restricted to the $\exists$ conditions ($\chi(1) = 5.0012$, $p = 0.025$).²

This refutes the hypothesis that the results of experiment I are entirely due to the mere presence of negation. Negation in the second conjunct truly affects pronoun interpretation, specifically in licensing a universal reading otherwise absent from a sentence without this negation.

4. Discussion and conclusion

Overall, the results of the experiments support the claim that the readings of cross-conjunction anaphora are sensitive to the presence of negation. A sentence like (11a) only seems to receive an existential reading. When negation is introduced in the clause that contains the pronoun, as in (11b), both an existential and a universal reading are possible.

- (11) a. There is a circle and it is blue.

²We fitted a CLM mixed-effects model with both factors and an interaction term, and compared it with a purely additive model without the interaction term, each time with a random intercept for participants.

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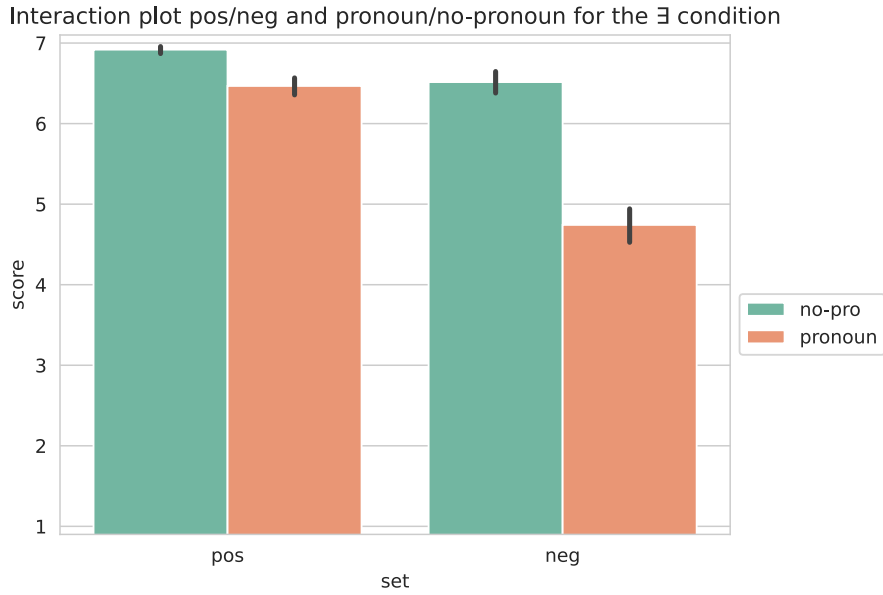


Figure 5: Bar plot of mean scores for POS/NEG- $\exists$ conditions in experiment I (pronoun) and experiment II (no-pronoun)

- b. There is a circle and it is not blue.

This result is theoretically interesting in more than one respect. First, it extends a generalization proposed by Kanazawa (1994) for the interpretation of donkey sentences like (12). Second, it raises a challenge for many theories in the Dynamic tradition that don't expect negation to affect the interpretation received. Let us explore these consequences in turn.

- (12) No farmer who owns a donkey cherishes it. ()
- existential:** ... *cherishes any donkey they own*
 - universal:** ... *cherishes every donkey they own*

4.1. Kanazawa's generalization

Kanazawa notes that there is a link between the monotonicity properties of the quantifier used in donkey sentences (Q in (13)) and the preferred reading of that sentence. For instance, it has been observed that (12) is preferentially read with the existential reading in (12a). Building on previous literature, Kanazawa also thought that each quantifier Q came with a preferred reading as outlined in (13).

- (13) Q farmer who owns a donkey cherishes it. ()
- $Q = \text{Every}$: preferred universal
 - $Q = \text{Some}$: preferred existential
 - $Q = \text{No}$: preferred existential

He proposed to summarize the data as follows:

- (14) **Kanazawa's generalization**³: In the default case, the only possible interpretation of

³Typically, Kanazawa's generalization refers to another generalization (presented later in the article) which states that the preferred reading is the one that preserves the monotonicity properties of the quantifier. Interestingly,

a donkey sentence whose determiner has the same monotonicity in its restrictor and scope argument is the existential reading, while the default interpretation of a donkey sentence whose determiner has different monotonicity properties in its restrictor and its scope is the universal reading.

Evidence from later experimental studies Foppolo (2008); Denić and Sudo (2022); Sun et al. (2020) suggest an amendment to Kanazawa’s facts is needed. These studies typically find that, in the case of *every*, there is no clear preference for a universal reading. In experiment I of Foppolo (2008) for instance, the critical condition where the existential reading is true and the universal reading false is accepted 57% of the time⁴. In experiment I of Sun et al. (2020), the critical condition is accepted 63%. In experiment 3A of Denić and Sudo (2022), it is accepted 52.4%.

On the other hand, the same studies all reveal a large preference for the existential reading when the quantifier is *no* or *some*. In Foppolo (2008), the critical condition was accepted 7% in the case of *no* and 13% in the case of *some*. Similar results are found for *some* and *no* in Sun et al. (2020) (although they are able to bring out the universal reading for certain existential quantifiers in act-out tasks).

Overall, the existential reading always seems available regardless of quantifier. The universal reading becomes possible (in the default case) when the quantifier is *every* and the resulting interpretation is monotonic in the way described by Kanazawa. This leads to the following revision to Kanazawa (1994)’s generalization:

- (15) **Kanazawa’s generalization** (revised): *In the default case, an existential interpretation is available. When the determiner has different monotonicity properties in its restrictor and scope, a universal reading is also available.*

Kanazawa’s generalization (original or revised) is only a generalization about donkey sentences. However, we find that the amended generalization in (15) is strikingly consistent with our result on cross-conjunction anaphora. When no negation is present, the pronoun and its antecedent both stand in environments of the same monotonicity (upward monotone) and, concomitantly, the resulting reading is existential. When negation is introduced, the pronoun can plausibly take scope under negation. If it does, the pronoun and its antecedent stand in environments of different monotonicity and the universal reading becomes available.⁵

this generalization, even when amended as we suggest below, does not extend to our data about cross-conjunction anaphora: the universal reading, even when the second conjunct contains negation, does not preserve monotonicity.

⁴This is only reporting truth/falsity judgments. Foppolo argues that there is a contrast between the critical condition that I described and another condition that makes both readings true in terms of both reaction times and acceptability judgments (which participants had to provide if they provided true answers). But this could simply reflect the fact that, in the critical condition, the two readings yield different truth value forcing participants to make a definite choice of interpretation. It is not evidence that the existential reading is dispreferred.

⁵What we call the *universal reading* corresponds to a paraphrase in which a pronoun is replaced by a universal quantifier scoping over negation: “There is a circle, and every circle has the property of being not blue.” Of course, this is logically equivalent to an *existential* paraphrase under negation: “There is a circle, and it is not the case that a circle is blue.” What matters for this discussion, however, is simply that if the pronoun takes scope under negation, the monotonicity of its environment is reversed compared to that of the indefinite. Kanazawa’s generalization thus leads us to expect that *two readings* will be available, instead of just one in the absence of negation. This is precisely what we observe, regardless of how the two readings are characterized – whether as ‘universal’ or ‘existential’.

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- (16) a. There is a circle and it is blue. (existential, universal)
b. There is a circle and it is not blue. (existential, universal)

4.2. Dynamic theories

While in line with Kanazawa's generalization, our results nevertheless contradict Egli's principle. Egli's principle is a consequence of many theories in the dynamic tradition, including those that do justice to Kanazawa's generalization in the donkey case Champollion et al. (2019). It is given in (17). Informally put, this principle states that an existential can 'extend' its scope over a whole conjunction.

$$(17) \quad [\exists x, p] \wedge q \Leftrightarrow \exists x, [p \wedge q]$$

Concretely, this principle expects that the a and b sentences (18) are rigorously synonymous with the a and b sentences in (19) respectively.

- (18) a. There is a circle and it is blue.
b. There is a circle and it is not blue.
- (19) a. There is a circle that is blue.
b. There is a circle that is not blue.

However, the explicit comparison that we ran in experiment 2 (section 3) shows that this is not the case. In the critical condition where the display contains a blue and a red circle, (18b) is rated the lowest.

4.3. Towards a resolution

We briefly sketch two paths towards an analysis. First, one could posit that conjunction is ambiguous between existential and universal readings. In dynamic semantics, it is possible to construct dynamic denotations for conjunction that generate universal readings (for which Egli's principle is not valid). This can be done by exploiting e.g. the classical equivalence in (20), and reinterpreting the material conditional ' $\rightarrow$ ' as its dynamic equivalent.

$$(20) \quad A \wedge B \equiv \neg[A \rightarrow \neg B]$$

Such a move would predict that both (18a) and (18b) are ambiguous between an existential and a universal reading. This is not what we observed: only the existential reading is found in (18b). To predict our observations, the theory would need to be supplemented with a principle restricting the universal reading in the positive case in (18a).

This is a difficult task. However, the idea that universal readings are marginally available even in positive contexts has been defended in previous literature (van der Does, 1993; Chatain, 2018). Chatain (2018) provides the example in (21) (adapting similar examples known from donkey sentences, e.g. Chierchia (1988) and previous literature). He argues that the bet is lost if Jane has two umbrellas and brought one of them to work.

- (21) I bet \$10 that Jane has an umbrella and that she left it at home today.

While the evidence in (21) is encouraging, it remains to be explained in this line of analysis why there is a contrast between the positive and the negative case. In the negative case, the ambiguity could be revealed even in the decontextualized sentences proposed in our experiments.

In the positive case, our experiment could not detect this ambiguity. Achieving a universal interpretation in positive contexts appears to crucially depend on specific contextual and lexical adjustments, such as those used in (21).

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